

1. METRIC SPACES

Definition 1.1 (Metric). A *metric* is a function defined by

$$(1) \quad d : M \times M \rightarrow \mathbb{R}$$

such that for any $x, y, z \in M$, the separation axiom

$$(2) \quad d(x, y) \geq 0,$$

the indiscernible identity

$$(3) \quad d(x, y) = 0 \iff x = y$$

the symmetry axiom

$$(4) \quad d(x, y) = d(y, x)$$

and the triangle inequality

$$(5) \quad d(x, z) \leq d(x, y) + d(y, z)$$

all hold. A *metric space* is a set with a metric defined.

Definition 1.2 (Absolute Difference). The *absolute difference* is a function $\mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$

$$(6) \quad d(x, y) := |x - y|$$

Definition 1.3 (Euclidean Metric).

$$(7) \quad d(x, y) := \sqrt{\sum_i d_i(x_i, y_i)^2}$$

Definition 1.4 (Taxi-Cab Metric).

$$(8) \quad d(x, y) := \sum_i d_i(x_i, y_i)$$

Definition 1.5 (Maximum Metric).

$$(9) \quad d(x, y) := \max\{d_i(x_i, y_i)\}$$

2. BOUNDED SEQUENCES

Definition 2.1 (Bounded Sequence). A sequence is *bounded* if there is a $p \in M$ and $r \in \mathbb{R}$, such that

$$(10) \quad \forall n \in \mathbb{N} : d(s_n, p) \leq r$$

Definition 2.2 (Pinched Sequence). A sequence is *pinched* by $r \in \mathbb{R}$, if

$$(11) \quad \forall m, n \in \mathbb{N} : d(s_m, s_n) \leq r$$

Definition 2.3. A real sequence is *bounded above* by B if

$$(12) \quad \forall n : s_n < B$$

and is *bounded below* by b if

$$(13) \quad \forall n : b < s_n$$

A sequence is *totally bounded* if it is bounded above and below.

Theorem 2.4. A real sequence is *totally bounded*, iff it is *bounded*.

Proof. Let $s : \mathbb{N} \rightarrow \mathbb{R}$ be a totally bounded sequence

$$(14) \quad \forall n \in \mathbb{N} : b < s_n < B$$

Let p be the average of the two bounds

$$(15) \quad p := \frac{B+b}{2}$$

and r be the distance from B to p

$$(16) \quad r := B - p$$

Rearranging

$$B = r + pb = 2p - B$$

$$2p - (r + p) < s_n < r + p$$

$$p - r < s_n < p + r$$

$$|s_n - p| < r$$

Now assume that $\langle s_n \rangle$ is bounded, that is that there are $p, r \in \mathbb{R}$ such that

$$|s_n - p| < r$$

$$p - r < s_n < p + r$$

Let $b := p - r$ and $B := p + r$

$$(17) \quad b < s_n < B$$

□

3. CONVERGENT SEQUENCES

Definition 3.1 (Convergent Sequence). A sequence $\langle s_n \rangle$ in M *converges* to $l \in M$, if

$$(18) \quad \forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall n > N : d(s_n, l) < \varepsilon$$

Theorem 3.2. *Every convergent sequence is bounded.*

Proof. Let $M = (A, d)$ be a metric space and let $\langle s_n \rangle$ be a sequence in M that converges to $l \in M$. Let $\varepsilon > 0$. Since $\langle s_n \rangle$ converges, there is an N such that if $n > N$

$$(19) \quad d(s_n, l) < \varepsilon$$

Set B' be a bound of the sequence before N

$$B' := \max\{d(s_n, l) : n \leq N\}$$

Finally set

$$(20) \quad B := \max\{B', \varepsilon\}$$

Then B is a bound for all $n \in \mathbb{N}$

$$(21) \quad \forall n \in \mathbb{N} : d(s_n, l) \leq B$$

which implies that $\langle s_n \rangle$ is bounded. □

Theorem 3.3. *If*

$$(22) \quad \begin{aligned} S &= \lim_n s_n \\ T &= \lim_n t_n \end{aligned}$$

Then

$$(23) \quad \lim_n (s_n + t_n) = S + T$$

Proof. Choose an $\varepsilon' > 0$ and define $\varepsilon := 2\varepsilon'$. By the definition of convergence, there exists an N_s and N_t such that

$$(24) \quad n > N_s \Rightarrow |s_n - S| < \varepsilon'$$

$$(25) \quad n > N_t \Rightarrow |t_n - T| < \varepsilon'$$

Let $N := \max\{N_s, N_t\}$, then for $n > N$, by the triangle inequality

$$(26) \quad |(s_n + t_n) - (S + T)| \leq |s_n - S| + |t_n - T|$$

By the definition of convergence

$$(27) \quad |s_n - S| + |t_n - T| < \varepsilon' + \varepsilon' = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Thus

$$(28) \quad \lim_n (s_n + t_n) = S + T$$

□

Theorem 3.4. *If*

$$(29) \quad \begin{aligned} S &= \lim_n s_n \\ T &= \lim_n t_n \end{aligned}$$

Then

$$(30) \quad \lim_n (s_n \cdot t_n) = ST$$

Proof. Choose an $\varepsilon' > 0$ By the definition of convergence, there exists an N_s and N_t such that

$$(31) \quad n > N_s \Rightarrow |s_n - S| < \varepsilon'$$

$$(32) \quad n > N_t \Rightarrow |t_n - T| < \varepsilon'$$

Let $N := \max\{N_s, N_t\}$, then for $n > N$

$$(33) \quad |(s_n t_n) - (ST)| = |s_n t_n - s_n T + s_n T - ST|$$

$$(34) \quad = |s_n(t_n - T) + (s_n - S)T|$$

By the triangle inequality

$$(35) \quad \begin{aligned} |s_n(t_n - T) + (s_n - S)T| &\leq |s_n(t_n - T)| + |(s_n - S)T| \\ &\leq |s_n||t_n - T| + |s_n - S||T| \end{aligned}$$

Since every convergent sequence is also bounded, there is some bound B , such that for every n

$$(36) \quad |s_n| < B$$

$$(37) \quad |s_n||t_n - T| + |s_n - S||T| \leq B|t_n - T| + |s_n - S||T|$$

Let $M := \max\{C, |T|\}$. Then

$$(38) \quad B|t_n - T| + |s_n - S||T| \leq M(|s_n - S| + |t_n - T|)$$

Let $\varepsilon := 2M\varepsilon'$, then

$$(39) \quad |s_n - S| < \varepsilon' = \frac{\varepsilon}{2M}$$

$$(40) \quad |t_n - T| < \varepsilon' = \frac{\varepsilon}{2M}$$

Then

$$(41) \quad M(|s_n - S| + |t_n - T|) < M\left(\frac{\varepsilon}{2M} + \frac{\varepsilon}{2M}\right) = \varepsilon$$

Thus, for $n > N$

$$(42) \quad |s_n t_n - ST| < \varepsilon$$

or

$$(43) \quad \lim_n s_n t_n = ST$$

□

Theorem 3.5. *If a sequence converges, then the limit is unique.*

Proof. Assume without loss of generality that some convergent sequence s has two distinct limits L and L' . Define the distance between these two limits $d := |L - L'|$. From the definition of the metric, $d > 0$. From the definition of convergence, there is a N and N' such that

$$(44) \quad n > N \Rightarrow |s_n - L| < \frac{d}{2}$$

and

$$(45) \quad n > N' \Rightarrow |s_n - L'| < \frac{d}{2}$$

Let $M := \max\{N, N'\}$. For $n > M$

$$(46) \quad d := |L - L'| = |L - s_n + s_n - L'|$$

By the triangle identity

$$(47) \quad |L - s_n + s_n - L'| \leq |s_n - L| + |s_n - L'|$$

From the definition of convergence

$$(48) \quad |s_n - L| < \frac{d}{2}$$

$$(49) \quad |s_n - L'| < \frac{d}{2}$$

By $a < b \wedge c < d \Rightarrow a + c < b + d$

$$(50) \quad |s_n - L| + |s_n - L'| < \frac{d}{2} + \frac{d}{2} = d$$

which implies

$$(51) \quad d < d$$

which is a contradiction. □

Theorem 3.6 (Squeeze Theorem). *For all sequences s , and all convergent sequences r, t such that*

$$(52) \quad \begin{aligned} \lim_n r_n &= L \\ \lim_n t_n &= L \end{aligned}$$

and for all $n \in \mathbb{Z}^+$

$$(53) \quad r_n < s_n < t_n$$

implies that s is a convergent sequence and its limit is given by

$$(54) \quad \lim_n s_n = L$$

Proof. Fix $\varepsilon > 0$. Then by the definition of the convergent sequence

$$(55) \quad \begin{aligned} \exists N_1 : n > N_1 &\Rightarrow |y_n - L| < \varepsilon \\ \exists N_2 : n > N_2 &\Rightarrow |z_n - L| < \varepsilon \end{aligned}$$

Let $N := \max\{N_1, N_2\}$, then for all $n > N$

$$(56) \quad \begin{aligned} L - \varepsilon &< y_n < L + \varepsilon \\ L - \varepsilon &< z_n < L + \varepsilon \end{aligned}$$

Using the trichotomy property of order

$$(57) \quad L - \varepsilon < y_n < x_n < z_n < L + \varepsilon$$

or

$$(58) \quad L - \varepsilon < x_n < L + \varepsilon$$

Therefor $n > N$ implies

$$(59) \quad |x_n - L| < \varepsilon$$

or

$$(60) \quad \lim_n x_n = L$$

□

Theorem 3.7.

$$(61) \quad \lim_n \frac{1}{n} = 0$$

Proof. Choose an $\varepsilon' \in \mathbb{R}^+$. By the Archimedean property

$$(62) \quad \exists N \in \mathbb{Z}^+ : N > \varepsilon'$$

Since both $N > 0$ and $\varepsilon' > 0$

$$(63) \quad \frac{1}{N} < \frac{1}{\varepsilon'}$$

Define $\varepsilon := \frac{1}{\varepsilon'} \in \mathbb{R}^+$

$$(64) \quad \frac{1}{N} < \varepsilon$$

Let $n > N > 0$, then

$$(65) \quad \frac{1}{N} > \frac{1}{n}$$

By trichotomy

$$(66) \quad \frac{1}{n} < \varepsilon$$

$$(67) \quad n > N \Rightarrow \left| \frac{1}{n} - 0 \right| < \varepsilon$$

or

$$(68) \quad \lim_n \frac{1}{n} = 0$$

□

Lemma 3.8. *If $\{|a_n - L|\}$ converges to zero, then $\{a_n\}$ converges to L .*

$$(69) \quad \lim_n |a_n - L| = 0 \Rightarrow \lim_n a_n = L$$

Proof. Since $\{|a_n - L|\}$ converges to zero, for any $\varepsilon > 0$, there is an N

$$(70) \quad n > N \Rightarrow ||a_n - L| - 0| < \varepsilon$$

$$(71) \quad ||a_n - L| < \varepsilon$$

$$(72) \quad n > N \Rightarrow |a_n - L| < \varepsilon$$

Then

$$(73) \quad \lim_n a_n = L$$

□

4. CLOSED SETS

Definition 4.1 (Closed Set). A subset $S \subset M$ is *closed* in M if every sequence $\langle s_n \rangle$ in S that converges in M also converges in S .

$$(74) \quad \lim_n s_n \in M \Rightarrow \lim_n s_n \in S$$

Theorem 4.2. *The real interval $[a, b]$ is closed.*

$$(75) \quad \forall s : \mathbb{N} \rightarrow [a, b] : \lim_n s_n = L \Rightarrow L \in [a, b]$$

Proof. Suppose not, then

$$(76) \quad \exists s : \mathbb{N} \rightarrow [a, b] : \lim_n s_n = L \wedge (L < a \vee b < L)$$

From the definition of a limit

$$(77) \quad \forall \varepsilon > 0 : \exists N \in \mathbb{N} : n > N \Rightarrow L - \varepsilon < s_n < L + \varepsilon$$

Case 1 ($b < L$). Then $L - b > 0$. Let $\varepsilon = L - b$, then there is a N_1 , such that $n > N_1$

$$(78) \quad L - L + b = b < s_n$$

but the definition of the sequence restricts the domain to $s_n < b$, so we obtain a contradiction.

Case 2 ($a < L$). Then $a - L > 0$, let $\varepsilon = a - L$, which implies there is a N_2 , such that $n > N_2$

$$(79) \quad s_n < L + a - L = a$$

which contradicts the restriction on the domain $a < s_n$

□

Theorem 4.3 (Inheritance Principle). *If S is closed in M and N is a subspace of M , then $N \cap S$ is closed in N .*

Proof. Let N be a subspace of M and S some closed subset of M . Define a sequence $s : \mathbb{N} \cap S$ such that

$$(80) \quad \exists L \in N : \forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall n \in \mathbb{N} : |s_{N+n} - L| < \varepsilon$$

Since S is closed and $N \cap M$, $L \in S \cap N$. Hence $S \cap N$ is closed. □

Corollary 4.4. *If N is closed in M , then S is closed in N if and only if S is closed in M .*

5. MONOTONIC SEQUENCES

Definition 5.1 (Monotonic Sequence). An *increasing sequence* is a sequence where

$$(81) \quad m > n \Rightarrow s_m > s_n$$

A *strictly increasing sequence* is a sequence where

$$(82) \quad m > n \Rightarrow s_m \geq s_n$$

Similarly, a *decreasing sequence* is a sequence where

$$(83) \quad m > n \Rightarrow s_m < s_n$$

And a *strictly decreasing sequence* is a sequence where

$$(84) \quad m > n \Rightarrow s_m \leq s_n$$

A *monotonic sequence* is a sequence that is either increasing or decreasing.

Lemma 5.2. *If $n : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ is a strictly increasing sequence ($i > j \Rightarrow n_i > n_j$), then for any positive integer N*

$$(85) \quad \exists i \in \mathbb{Z}^+ : n_i > N$$

Proof. Hypothesis for induction:

$$(86) \quad \forall i \in \mathbb{Z}^+ : n_i \geq i$$

Case 1 (Base). If $n_1 \in \mathbb{Z}^+$, then $n_1 \geq 1$.

Case 2 (Induction). Assume without loss of generality,

$$(87) \quad n_i \geq i$$

Since n is strictly increasing

$$(88) \quad n_{i+1} > n_i$$

By trichotomy

$$(89) \quad n_{i+1} > i$$

Then

$$(90) \quad n_{i+1} \geq i + 1$$

Thus by induction,

$$(91) \quad \forall i \in \mathbb{Z}^+ : n_i \geq i$$

□

6. SUBSEQUENTIAL LIMITS

Definition 6.1 (Subsequence). A *subsequence* of a sequence s is a sequence that is a composition $s \circ n$ where $n : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ is strictly increasing.

Definition 6.2 (Tail). The N -tail of a sequence $s : n \mapsto s_n$ is the subsequence s_{N+n} , where $N \in \mathbb{Z}^{\geq 0}$

Theorem 6.3. *If a sequence converges to a limit, then all of its subsequential limits are equal to that limit.*

Proof. Consider a sequence s , that converges to L , and $s \circ n$ a subsequence of s . Fix $\varepsilon > 0$. Since s converges to L , there exists an N such that

$$(92) \quad n > N \Rightarrow |s_n - L| < \varepsilon$$

Let $I := \inf\{i \in \mathbb{Z}^+ : n_i > N\}$ (The set is not empty, by the lemma above $n_N \geq N$). Since $n_I \geq I > N$

$$(93) \quad n > n_I \Rightarrow |s_n - L| < \varepsilon$$

From the definition of strictly increasing

$$(94) \quad i > I \Rightarrow n_i > n_I$$

So

$$(95) \quad i > I \Rightarrow n_i > n_I \Rightarrow |s_{n_i} - L| < \varepsilon$$

Or

$$(96) \quad \lim_i s_{n_i} = L$$

□

Theorem 6.4. *If a sequence is bounded, then all subsequences have the same bounds.*

Proof. Let s be a sequence bounded by (b, B)

$$(97) \quad \forall n : b \leq s_n \leq B$$

Let $s \circ n$ be some subsequence, then

$$(98) \quad \forall i : b \leq s_{n_i} \leq B$$

□

Theorem 6.5. *Every subsequence of a monotonic sequence is the same type of monotonic.*

Proof. Let s be a sequence and $\sim$ be some order relation on the monotonic sequence such that

$$(99) \quad m > n \Rightarrow s_m \sim s_n$$

Let $s \circ n$ be some subsequence, then

$$(100) \quad i > j \Rightarrow n_i > n_j \Rightarrow s_{n_i} \sim s_{n_j}$$

□

Lemma 6.6. *If a tail of a sequence has a maximum, the parent sequence has a maximum.*

Proof. Let $a := \max\{s_n : n > N\}$, which exists by assumption. Let $b := \max\{s_n : n \leq N\}$, which exists because the set is finite. If we let $c := \max\{a, b\}$, then $c = \max\{s_n : n \in \mathbb{Z}^+\}$. $\square$

Theorem 6.7 (Point Peak Theorem). *Every real sequence has a monotonic subsequence.*

Proof. Let $\langle s_n \rangle$ be any real sequence. Either every tail of $\langle s_n \rangle$ has a maximum

$$(101) \quad \forall N \in \mathbb{Z}^{\geq 0}, \exists m, \forall n : s_{N+m} \geq s_{N+n}$$

or there is a tail without a maximum.

$$(102) \quad \exists N \in \mathbb{Z}^{\geq 0}, \forall m, \exists n : s_{N+m} < s_{N+n}$$

First suppose every tail of $\langle s \rangle$ has a maximum

$$\forall N \in \mathbb{Z}^{\geq 0}, \exists m, \forall n : s_{N+m} \geq s_{N+n}$$

Define a sequence $k : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$

$$(103) \quad k_1 := \min\{m | \forall n : s_m \geq s_n\}$$

$$(104) \quad k_{i+1} := \min\{m | \forall n : s_{k_i+m} \geq s_{k_i+n}\} + k_i$$

The sequence $s \circ k$ is a subsequence because k strictly increasing

$$(105) \quad m \in \mathbb{Z}^+ \Rightarrow k_{i+1} > k_i$$

From the definition of k_{i+1}

$$(106) \quad \forall i, n : s_{k_{i+1}} \geq s_{k_i+n}$$

Since k is strictly increasing

$$(107) \quad k_{i+2} > k_{i+1} > k_i$$

Let n be the positive integer such that

$$(108) \quad n := k_{i+2} - k_i \in \mathbb{Z}^+$$

Applying this n to the definition

$$(109) \quad s_{k_{i+1}} \geq s_{k_i+k_{i+2}-k_i} = s_{k_{i+2}}$$

which shows that $s \circ k$ is decreasing.

$$(110) \quad s_{k_i} \geq s_{k_{i+1}}$$

Now suppose there is a tail that does not have a maximum

$$\exists N \in \mathbb{Z}^{\geq 0}, \forall m, \exists n : s_{N+m} < s_{N+n}$$

Every tail of the tail without a maximum has no maximum, by lemma 6.6.

$$(111) \quad \forall M \in \mathbb{Z}^{\geq 0}, m, \exists n : s_{N+M+m} < s_{N+M+n}$$

This implies there is always a larger number further down the sequence.

$$(112) \quad \forall m, \exists n : s_{N+m} < s_{N+m+n}$$

Define a sequence $k : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$

$$(113) \quad k_1 := N + 1$$

$$(114) \quad k_{i+1} := \min\{n : s_{k_i+n} > s_{k_i}\} + k_i$$

The sequence $s \circ k$ is a subsequence because k is strictly increasing

$$(115) \quad n \in \mathbb{Z}^+ \Rightarrow k_{i+1} > k_i$$

From the definition of k_{i+1} , $s \circ k$ is strictly increasing

$$(116) \quad s_{k_{i+1}} > s_{k_i}$$

Thus all real sequences have monotonic subsequences. $\square$

Theorem 6.8. For $|a| < 1$

$$(117) \quad \lim_n a^n = 0$$

Proof. Let a be a number such that $|a| < 1$. Since $|a| > 0$, multiplying the inequalities by $|a|$ gives

$$(118) \quad 0 \geq |a|^2 < |a|$$

$$(119) \quad 0 \geq |a^2| < |a|$$

Assume without loss of generality

$$(120) \quad 0 \geq |a^{n+1}| < |a^n|$$

Then multiplying by $|a|$ gives

$$(121) \quad 0 \geq |a^{n+1}| |a| < |a^n| |a|$$

$$(122) \quad 0 \geq |a^{n+2}| < |a^{n+1}|$$

Then, by induction, for all $n \in \mathbb{Z}^+$

$$(123) \quad 0 \geq |a^{n+1}| < |a^n|$$

which implies that the sequence $|a^n|$ is strictly decreasing, and bounded below by zero. By the monotonic convergence theorem, $|a^n|$ converges. Assume it converges to limit L .

$$(124) \quad L := \lim_n |a^n|$$

The sequence $|a^{2n}|$ is a subsequence of $|a^n|$, so it converges to the same limit

$$(125) \quad L = \lim_n |a^{2n}|$$

$$(126) \quad = \lim_n |a^n|^2$$

$$(127) \quad = \left(\lim_n |a^n| \right)^2$$

$$(128) \quad = L^2$$

Thus, either $L = 0$ or $L = 1$. Since $|a^1| = |a| < 1$ and a^n is strictly decreasing, $L \neq 1$. Therefore

$$(129) \quad \lim_n |a^n| = 0$$

By lemma 3.8

$$(130) \quad \lim_n a^n = 0$$

$\square$

7. COMPACT SETS

Definition 7.1 (Compact Set). A set is *compact* if every sequence $s : \mathbb{Z}^+ \rightarrow S$ has a subsequential limit in S .

Theorem 7.2. *A compact subset of M is a closed subset of M*

Proof. Let S be a compact subset of M and $s : \mathbb{N} \rightarrow S$ be a sequence that converges to some $L \in M$. Since s converges, every subsequential limit is equal to L . Since S is compact $L \in S$. Thus S is closed. $\square$

Theorem 7.3. *A compact subset of M is bounded in M*

Proof. Let p be a point in M and $S \subset M$. Suppose S is not bounded, then there is some sequence $s : \mathbb{N} \rightarrow S$ defined by

$$(131) \quad d(s_n, p) \geq n$$

Since S is compact, there is some subsequence of s that converges. For every $\varepsilon > 0$, there is an N

$$(132) \quad i > N \Rightarrow d(s_{n_i}, L) < \varepsilon$$

Choose some $\varepsilon > 0$. By the Archimedean property there is an N' such that

$$(133) \quad \varepsilon < N'$$

Let $M := \max\{N, N'\}$. Since $M > N$

$$(134) \quad d(s_{n_M}, p) < \varepsilon < N' < M$$

And since $s \circ n$ is a subsequence $M \leq n_M$, which along with the definition of the sequence gives

$$(135) \quad M \leq n_M \leq d(s_{n_M}, p) < M$$

which is a contradiction. So S is bounded. $\square$

8. FUNDAMENTAL SEQUENCES

Definition 8.1 (Fundamental Sequence). A sequence is *fundamental* if every ε has a N -tail, that is ε -pinched.

$$(136) \quad \forall \varepsilon > 0 : \exists N \in \mathbb{N} : \forall m, n > N : d(s_m, s_n) < \varepsilon$$

Theorem 8.2. *Every convergent sequence is fundamental.*

9. COMPLETE SPACES

Definition 9.1. A space is *complete* if every fundamental sequence is convergent.

Theorem 9.2 (Monotone Convergence Theorem). *A monotonic sequence converges if and only if it is bounded.*

Proof. Let s be an increasing sequence bounded above by B .

$$(137) \quad m > n \Rightarrow B \geq s_m \geq s_n$$

Without loss of generality, assume s is non-fundamental

$$(138) \quad \exists \varepsilon > 0, \forall N : \exists m, n > N : |s_m - s_n| > \varepsilon$$

Let $m > n$ (If $m = n$, then $s_m - s_n = 0 \not> \varepsilon$).

$$(139) \quad \forall n : \exists m > n : |s_m - s_n| > \varepsilon$$

Since s is increasing

$$(140) \quad \forall n : \exists m > n \Rightarrow s_m - s_n > \varepsilon$$

$$(141) \quad s_m > \varepsilon + s_n$$

Let $p := m - n \Rightarrow m = k + n$

$$(142) \quad \forall n, \exists p : s_{n+p} > \varepsilon + s_n$$

Define a sequence $k : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$

$$(143) \quad k_1 := 1$$

$$(144) \quad k_{i+1} := \min\{p | s_{k_i+p} > s_{k_i} + \varepsilon\} + k_i$$

The sequence $s \circ k$ is a subsequence because k is strictly increasing

$$(145) \quad p \in \mathbb{Z}^+ \Rightarrow k_{i+1} > k_i$$

By construction of k

$$(146) \quad s_{k_{i+1}} > s_{k_i} + \varepsilon$$

Let $A := s_1 = s_{n_1}$. Consider the induction hypothesis

$$(147) \quad \forall i > 1 : s_{k_i} - A > (i-1)\varepsilon$$

Case 1 (Base ($i = 2$)).

$$(148) \quad s_{n_2} > s_{n_1} + \varepsilon = A + (2-1)\varepsilon$$

Case 2 (Induction). Assume without loss of generality

$$(149) \quad s_{k_i} > A + (i-1)\varepsilon$$

Along with

$$(150) \quad s_{k_{i+1}} - \varepsilon > s_{k_i}$$

Gives

$$(151) \quad s_{k_{i+1}} - \varepsilon > A + (i-1)\varepsilon$$

$$(152) \quad s_{k_{i+1}} - A > [(i+1)-1]\varepsilon$$

By induction

$$(153) \quad \forall i > 1 : s_{n_i} - A > (i-1)\varepsilon$$

Choose $N \in \mathbb{Z}^+$ such that

$$(154) \quad N > \frac{B-A}{\varepsilon} + 1$$

By the Archimedean property, N exists.

$$(155) \quad s_N - A > \left(\frac{B-A}{\varepsilon} + 1 - 1 \right) \varepsilon$$

$$(156) \quad s_N > B$$

Which contradicts the assumption that s is bounded above by B . Therefore if s is bounded above, it is fundamental. By the completion theorem, since s is fundamental, it is also convergent. Thus it has shown that bounded, increasing

sequences are convergent. If s is a bounded, monotonic decreasing sequence, then $-s$ is bounded, monotonic increasing sequence, and therefor convergent, which implies s is convergent. Thus all bounded, monotonic sequences are convergent. The converse is proven by theorem 3.2, all convergent sequences are bounded. $\square$

10. SUMMABLE SEQUENCES

Definition 10.1 (Series). A *series* of partial sums of a is the sequence

$$(157) \quad s_n := \sum_{i=1}^n a_i$$

Definition 10.2 (Summable Sequence). A sequence is *summable* if its series of partial sums converges

$$(158) \quad \sum_{i=1}^{\infty} a_i := \lim_n \sum_{i=1}^n a_i$$

The *sum* of a sequence is the limit of its series.

Theorem 10.3. *Each sequence with the same sign has a monotonic series.*

- *A non-negative sequence has an increasing series.*
- *A positive sequence has a strictly increasing series.*
- *A non-positive sequence has a decreasing series.*
- *A negative sequence has a strictly decreasing series.*

Proof. Let $\sim$ be the relationship between the sequence a and zero.

$$(159) \quad \forall n : a_n \sim 0$$

Then

$$(160) \quad a_{n+1} \sim 0$$

$$(161) \quad a_{n+1} + s_n \sim s_n$$

$$(162) \quad s_{n+1} \sim s_n$$

$\square$

Theorem 10.4 (Series Linearity). *If a and b are summable and c and d are some numbers, then $ca + bd$ is summable and*

$$(163) \quad c \sum_{i=1}^{\infty} a_i + d \sum_{i=1}^{\infty} b_i = \sum_{i=1}^{\infty} (ca_i + db_i)$$

Proof.

$$(164) \quad c \sum_{i=1}^{\infty} a_i + d \sum_{i=1}^{\infty} b_i = c \lim_n \sum_{i=1}^n a_i + d \lim_n \sum_{i=1}^n b_i$$

$$(165) \quad = \lim_n \left(c \sum_{i=1}^n a_i + d \sum_{i=1}^n b_i \right)$$

$$(166) \quad = \lim_n \sum_{i=1}^n (ca_i + db_i)$$

$$(167) \quad = \sum_{i=1}^{\infty} (ca_i + db_i)$$

□

Theorem 10.5 (Fundamental Criterion). *A sequence a is summable if and only if*

$$(168) \quad \forall \varepsilon > 0 : \exists N : n, m > N : \left| \sum_{i=m+1}^n a_i \right| < \varepsilon$$

Proof. Fix $\varepsilon > 0$. By assumption, there is an N such that

$$(169) \quad n, m > N \Rightarrow \left| \sum_{i=m+1}^n a_i \right| < \varepsilon$$

If $n \leq m$, $\sum_{i=m+1}^n a_i = 0$, which is less than all $\varepsilon > 0$. Restrict $n > m$, then

$$(170) \quad \left| \sum_{i=1}^n a_i - \sum_{i=1}^m a_i \right| < \varepsilon$$

which by the completeness theorem, shows that the series of a converges and a is summable. □

Theorem 10.6 (Vanishing Condition). *If a is summable*

$$(171) \quad \lim_n a_n = 0$$

Proof. Since a is summable it has a sum. Let S be the sum of a

$$(172) \quad S := \sum_{i=1}^{\infty} a_i$$

Then

$$(173) \quad \lim_n a_n = \lim_n \left(\sum_{i=1}^n a_i - \sum_{i=1}^{n-1} a_i \right)$$

$$(174) \quad = \lim_n \sum_{i=1}^n a_i - \lim_n \sum_{i=1}^{n-1} a_i$$

$$(175) \quad = S - S = 0$$

□

Theorem 10.7 (Boundness Criterion). *A non-negative sequence is summable, if and only if its series is bounded.*

Proof. Let a be a non-negative sequence. Then its series is increasing. By the monotonic convergence theorem, this monotonic series is convergent if and only if it is bounded. Therefor the non-negative sequence is summable if and only if it is bounded. □

Theorem 10.8 (Comparison Test). *If $\forall n : |a_n| \leq b_n$ and b is summable, then a is summable*

Proof. Choose $\varepsilon > 0$. By fundamental criterion, there is an N , such that $m, n > N$

$$(176) \quad \left| \sum_{i=m}^n b_i \right| < \varepsilon$$

Since b is non-negative

$$(177) \quad \sum_{i=m}^n b_i = \left| \sum_{i=m}^n b_i \right| < \varepsilon$$

Using the triangle inequality

$$(178) \quad \left| \sum_{i=m}^n a_i \right| \leq \sum_{i=m}^n |a_i| \leq \sum_{i=m}^n b_i < \varepsilon$$

By the fundamental criterion, a is then summable. $\square$

Theorem 10.9 (Geometric Series). *If $|a| < 1$, then $\langle a^k \rangle$ is summable. The sum is given by*

$$(179) \quad \sum_{i=0}^{\infty} a^i = \frac{1}{1-a}$$

Proof. Let $\langle s_n \rangle$ be the series of partial sums

$$(180) \quad s_n := \sum_{i=0}^n a^i$$

$$(181) \quad \lim_n s_n = \lim_n \frac{1 - a^{n+1}}{1 - a} = \frac{1}{1 - a}$$

$\square$

Lemma 10.10.

$$(182) \quad \limsup_n s_n < B \Rightarrow \exists N \in \mathbb{N} : \forall n \geq N : s_n \leq B$$

Proof. Let $\langle s_n \rangle$ be a bounded sequence

$$(183) \quad L := \limsup_n s_n$$

Define B such that

$$(184) \quad L < B$$

Suppose that

$$(185) \quad \forall N \in \mathbb{N} : \exists n > N : s_n > B$$

Define a sequence $k : \mathbb{N} \rightarrow \mathbb{N}$, such that

$$(186) \quad k_1 = \min\{n : s_n > B\}$$

$$(187) \quad k_{i+1} = \min\{n : s_{k_i+n} > B\} + k_i$$

Since $k_{i+1} > k_i$, $\langle s \circ k \rangle$ is a subsequence. By its definition, $\forall i \in \mathbb{N} : s_{k_i} > B > L$. Then $\lim_i s_{k_i} > \limsup_n s_n$ which is a contradiction. Therefor $\exists N : \forall n \geq N : s_n < B$ $\square$

Theorem 10.11 (Ratio Test). *If*

$$(188) \quad \limsup_n \left| \frac{a_{n+1}}{a_n} \right| < 1$$

Then a is summable.

Proof. Let a be some sequence such that

$$(189) \quad \limsup_n \left| \frac{a_{n+1}}{a_n} \right| < 1$$

Define s such that

$$(190) \quad \limsup_n \left| \frac{a_{n+1}}{a_n} \right| < s < 1$$

By the above lemma, there is an N , such that for every $n \geq N$

$$(191) \quad \left| \frac{a_{n+1}}{a_n} \right| < s$$

$$(192) \quad |a_{n+1}| < s|a_n|$$

Induction hypothesis

$$(193) \quad \forall k \in \mathbb{Z}^+ : |a_{N+k}| < s^k |a_N|$$

The base case is given by $n = N$

$$(194) \quad |a_{N+1}| < s|a_N|$$

Assume without loss of generality $|a_{N+k}| < s^k |a_N|$, multiplying by $s > 0$

$$(195) \quad |a_{N+k}|s < s^{k+1}|a_N|$$

Applying the inequality from before

$$(196) \quad |a_{N+k+1}| < |a_{N+k}|s < s^{k+1}|a_N|$$

Thus, by induction

$$(197) \quad \forall k \in \mathbb{Z}^+ : |a_{N+k}| < s^k |a_N|$$

By the comparison test, a_{N+k} is summable, therefor a_n is summable. $\square$

Definition 10.12 (Absolutely Summable). A sequence is *absolutely summable* if $\langle |a_n| \rangle$ is summable.

Theorem 10.13. *An absolutely summable sequence is summable.*

Proof. By assumption the series

$$(198) \quad \sum_{i=1}^{\infty} |a_i|$$

converges. Then by the fundamental criterion, for every $\varepsilon > 0$, there is an N , such that

$$(199) \quad m, n > N \Rightarrow \left| \sum_{i=m+1}^n |a_i| \right| < \varepsilon$$

$$(200) \quad \sum_{i=m+1}^n |a_i| \leq \left| \sum_{i=m+1}^n |a_i| \right| < \varepsilon$$

By the triangle identity

$$(201) \quad \left| \sum_{i=m+1}^n a_i \right| \leq \sum_{i=m+1}^n |a_i| < \varepsilon$$

Then, for every $\varepsilon > 0$, there is an N , such that

$$(202) \quad m, n > N \Rightarrow \left| \sum_{i=m+1}^n a_i \right|$$

By the fundamental criterion, a is summable. □

Lemma 10.14. *If $\langle a_n \rangle$ is non-negative and decreasing, then*

$$(203) \quad \forall m, n \in \mathbb{N} : \left| \sum_{i=m}^n (-1)^i a_i \right| \leq a_m$$

Proof. Fix $m \in \mathbb{N}$. Let the induction hypothesis be that for all $n \in \mathbb{N}$

$$(204) \quad 0 \leq \sum_{i=m}^n (-1)^{i-m} a_i \leq a_m$$

Case 1 (Basis for Induction $P(m)$). Since $\langle a_n \rangle$ is non-negative

$$(205) \quad 0 \leq a_m = (-1)^{m-m} a_m = \sum_{i=m}^m (-1)^{i-m} a_i$$

Case 2 (Basis for Induction $P(m+1)$). Since $\langle a_n \rangle$ is non-negative

$$\begin{aligned} a_m &\geq -a_{m+1} + a_m \geq 0 \\ a_m &\geq (-1)^{m+1-m} a_{m+1} + (-1)^{m-m} a_m \geq 0 \\ a_m &\geq \sum_{i=m}^{m+1} (-1)^{i-m} a_i \geq 0 \end{aligned}$$

For the induction step, suppose for $\forall k : m \leq k \leq n$

$$(206) \quad 0 \leq \sum_{i=m}^k (-1)^{i-m} a_i \leq a_m$$

Case 3. Further restrict $n-m$ to odd integers. Since $\langle a_n \rangle$ is non-negative $a_{n+1} \geq 0$ along with the induction hypothesis

$$(207) \quad 0 \leq a_{n+1} + \sum_{i=m}^n (-1)^{i-m} a_i$$

$$(208) \quad = (-1)^{n+1-m} a_{n+1} + \sum_{i=m}^n (-1)^{i-m} a_i$$

$$(209) \quad = \sum_{i=m}^{n+1} (-1)^{i-m} a_i$$

Since $\langle a_n \rangle$ is decreasing

$$(210) \quad a_{n+1} \leq a_n \Rightarrow a_{n+1} - a_n \leq 0$$

Along with the induction hypothesis

$$(211) \quad a_m \geq a_{n+1} - a_n + \sum_{i=m}^{n-1} (-1)^{i-m} a_i$$

$$(212) \quad = (-1)^{n+1-m} a_{n+1} + (-1)^{n-m} a_n + \sum_{i=m}^{n-1} (-1)^{i-m} a_i$$

$$(213) \quad = \sum_{i=m}^{n+1} (-1)^{i-m} a_i$$

Case 4. Now restrict $n - m$ to only even integers. Since $\langle a_n \rangle$ is decreasing

$$(214) \quad a_n \geq a_{n+1} \Rightarrow a_n - a_{n+1} \geq 0$$

Applying the induction hypothesis

$$(215) \quad 0 \leq -a_{n+1} + a_n + \sum_{i=m}^{-1} (-1)^{i-m} a_i$$

$$(216) \quad = (-1)^{n+1-m} a_{n+1} + (-1)^{n-m} a_n + \sum_{i=m}^{n-1} (-1)^{i-m} a_i$$

$$(217) \quad = \sum_{i=m}^{n+1} (-1)^{i-m} a_i$$

Since $\langle a_n \rangle$ is non-negative $-a_{n+1} \leq 0$. Applying the hypothesis gives

$$(218) \quad a_m \geq -a_{n+1} + \sum_{i=m}^n (-1)^{i-m} a_i$$

$$(219) \quad = (-1)^{n+1-m} a_{n+1} + \sum_{i=m}^n (-1)^{i-m} a_i$$

$$(220) \quad = \sum_{i=m}^{n+1} (-1)^{i-m} a_i$$

Therefor for all n , $P(n) \Rightarrow P(n+1)$. By induction, for all n ,

$$(221) \quad 0 \leq \sum_{i=m}^n (-1)^{i-m} a_i \leq a_m$$

Since the sum is greater than zero

$$(222) \quad \sum_{i=m}^n (-1)^{i-m} a_i = \left| \sum_{i=m}^n (-1)^{i-m} a_i \right|$$

$$(223) \quad = \left| (-1)^m \sum_{i=m}^n (-1)^{i-m} a_i \right|$$

$$(224) \quad = \left| \sum_{i=m}^n (-1)^i a_i \right| \leq a_m$$

□

Theorem 10.15 (Alternating Series Test). *If a is decreasing, non-negative and converges to zero, then $\langle (-1)^n a_n \rangle$ is summable.*

Proof. Let $\langle s_n \rangle$ be a series of partial sums

$$(225) \quad s_n := \sum_{i=0}^n (-1)^i a_i$$

Fix $\varepsilon > 0$. Since $\lim_n a_n = 0$, choose some N such that if $n > N$

$$(226) \quad |a_n| < \varepsilon$$

Choose n and m such that $n > m > N$, then

$$(227) \quad \left| \sum_{i=m}^n (-1)^i a_i \right| \leq a_m \leq |a_m| < \varepsilon$$

By the fundamental criterion, $\langle a_n \rangle$ is summable. $\square$

11. FUNCTIONAL LIMITS

Definition 11.1 (Limit). The *limit* L of a function $f : [a, b] \rightarrow \mathbb{R}$ at c is

$$(228) \quad \forall \varepsilon > 0 : \exists \delta > 0 : \forall x \in [A, B] : 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon$$

The limit at c of f is denoted by

$$(229) \quad \lim_{x \rightarrow c} f(x) = L$$

Theorem 11.2 (Sequential Limit Criterion).

$$(230) \quad \lim_{x \rightarrow c} f(x) = L$$

if and only if, for all sequences such that $\forall i : x_i \neq c$:

$$(231) \quad \lim_n x_n = c \Rightarrow \lim_n f(x_n) = L$$

Proof. Choose a sequence x such that $\forall x_n \neq c$ and

$$(232) \quad \lim_n x_n = c$$

Fix $\varepsilon > 0$. By assumption $\exists \delta > 0$ such that for every x

$$(233) \quad 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon$$

Since x_n converges to c , there is an N such that

$$(234) \quad n > N \Rightarrow |x_n - c| < \delta$$

Using the definition of the functional limit, this implies

$$(235) \quad |f(x_n) - L| < \varepsilon$$

Thus

$$(236) \quad \lim_n f(x_n) = L$$

For the converse, assume

$$(237) \quad \lim_{x \rightarrow c} f(x) \neq L$$

Then there is an $\varepsilon > 0$ such that for every $\delta > 0$ there is an x such that

$$(238) \quad 0 < |x - c| < \delta \wedge |f(x) - L| \geq \varepsilon$$

Choose such an ε . Generate a sequence such that

$$(239) \quad 0 < |x_n - c| < \frac{1}{n} \wedge |f(x_n) - L| \geq \varepsilon$$

By the squeeze theorem

$$(240) \quad \lim_n x_n = c$$

The other part of the generator

$$(241) \quad \forall n : |f(x_n) - L| \geq \varepsilon$$

implies

$$(242) \quad \lim_n f(x_n) \neq L$$

Thus

$$(243) \quad \lim_n x_n = c \Rightarrow \lim_n f(x_n) = L \Rightarrow \lim_{x \rightarrow c} f(x) = L$$

□

Theorem 11.3 (Squeeze Theorem). *If*

$$(244) \quad \lim_{x \rightarrow c} f(x) = L$$

$$(245) \quad \lim_{x \rightarrow c} g(x) = L$$

and

$$(246) \quad f(x) \leq h(x) \leq g(x)$$

Then

$$(247) \quad \lim_{x \rightarrow c} h(x) = L$$

Theorem 11.4 (Uniqueness). *If a function converges to a limit at some point, then that limit is unique.*

Theorem 11.5.

$$(248) \quad \lim_{x \rightarrow c} f(x) = F$$

$$(249) \quad \lim_{x \rightarrow c} g(x) = G$$

implies

$$(250) \quad \lim_{x \rightarrow c} [f(x) + g(x)] = F + G$$

Theorem 11.6.

$$(251) \quad \lim_{x \rightarrow c} f(x) = F$$

$$(252) \quad \lim_{x \rightarrow c} g(x) = G$$

implies

$$(253) \quad \lim_{x \rightarrow c} f(x)g(x) = FG$$

12. CONTINUOUS FUNCTIONS

Definition 12.1 (Continuous Function). A function f is continuous at a if

$$(254) \quad \lim_{x \rightarrow a} f(x) = f(a)$$

Theorem 12.2 (Sequential Limit Criterion). *For continuous functions f*

$$(255) \quad \lim_n f(x_n) = f\left(\lim_n x_n\right)$$

Theorem 12.3 (ε - δ Condition).

Theorem 12.4 (Closed Set Condition). *If $f : M \rightarrow N$ is continuous and M is closed, then the pre-image of each closed set in N is closed in M .*

Proof. Let I be a closed subset of N . Define a sequence $s : \mathbb{N} \rightarrow I \subset M$ such that

$$(256) \quad \lim_n s_n = L \in M$$

By sequential convergence condition

$$(257) \quad \lim_n f(s_n) = f(\lim_n s_n) = f(L)$$

Since the set I is closed

$$(258) \quad \lim_n f(s_n) \in I \Rightarrow f(L) \in I$$

So $L \in f^{-1}[I]$ which implies that $f^{-1}I$ is closed □

Theorem 12.5. *If $f : M \rightarrow N$ is continuous and I is compact in M , then the image of f restricted to I is compact*

Proof. Let $b : \mathbb{N} \rightarrow f[I]$ and define $a : \mathbb{N} \rightarrow I$ such that

$$(259) \quad f(a_n) = b_n$$

Since I is compact, a has a subsequential limit in I

$$(260) \quad L := \lim_i a_{n_i} \in I$$

Since f is continuous

$$(261) \quad \lim_i b_{n_i} = \lim_i f(a_{n_i}) = f(L) \in f[I]$$

There for the image of f is restricted to A is compact. □

Lemma 12.6.

$$(262) \quad a_{n+1} := \frac{a_n}{2} \Rightarrow a_n = \frac{1}{2^{n-1}} a_1$$

Proof. The induction hypothesis is that, $\forall n \in \mathbb{Z}^+$

$$(263) \quad a_n = \frac{1}{2^{n-1}} a_1$$

Case 1 (Base). The base case is trivially true

$$(264) \quad a_1 = \frac{1}{2^{1-1}} a_1$$

Case 2 (Induction). Assume without loss of generality

$$(265) \quad a_n = \frac{1}{2^{n-1}} a_1$$

From the opening assumption

$$(266) \quad a_{n+1} = \frac{a_n}{2}$$

$$(267) \quad = \frac{1}{2} \frac{1}{2^{n-1}} a_1$$

$$(268) \quad = \frac{1}{2^{n+1-1}} a_1$$

By induction, $\forall n \in \mathbb{Z}^+$

$$(269) \quad a_n = \frac{1}{2^{n-1}} a_1$$

□

Theorem 12.7 (Intermediate Value Theorem). *If $f : [A, B] \rightarrow \mathbb{R}$ is continuous and $f(A) \leq u \leq f(B)$ or $f(A) \geq u \geq f(B)$, then there exists $C \in [A, B] : f(C) = u$*

Proof. Define four sequences, a , b , c , and e with the following algorithm

$$(270) \quad a_1 := A$$

$$(271) \quad b_1 := B$$

$$(272) \quad c_n := \frac{1}{2}(a_n + b_n) e_n = b_n - a_n$$

If $f(c_n) < u$:

$$(273) \quad a_{n+1} = c_n$$

$$(274) \quad b_{n+1} = b_n$$

Else:

$$(275) \quad a_{n+1} = a_n$$

$$(276) \quad b_{n+1} = c_n$$

Using the algorithm, if $f(c_n) = u$, then there is a $C = c_n \in [A, B] : f(C) = u$, and we are done. For the rest of this proof, we will assume that $\forall n : f(c_n) \neq u$. Finishing off the algorithm, if $f(c_n) > u$

$$(277) \quad a_{n+1} = a_n$$

$$(278) \quad b_{n+1} = c_n = \frac{a_n + b_n}{2}$$

$$(279) \quad e_{n+1} = \frac{a_n + b_n}{2} - a_n = \frac{b_n - a_n}{2}$$

If $f(c_n) < u$

$$(280) \quad a_{n+1} = c_n = \frac{a_n + b_n}{2}$$

$$(281) \quad b_{n+1} = b_n$$

$$(282) \quad e_{n+1} = b_n - \frac{a_n + b_n}{2} = \frac{b_n - a_n}{2}$$

Since $\forall n : a_n < c_n < b_n$, a_n is increasing and b_n is decreasing. Since they are also bounded by (A, B) , by the monotone convergence theorem, they both converge. From the definition of e_n

$$(283) \quad e_n = b_n - a_n$$

So

$$(284) \quad e_{n+1} = \frac{e_n}{2}$$

By the lemma 12.6

$$(285) \quad e_n = \frac{1}{2^{n-1}} e_1 = \frac{1}{2^{n-1}} (B - A)$$

which converges to zero. Then $\lim_n b_n = \lim_n a_n$. Let $C := \lim_n a_n$, for continuous functions f

$$(286) \quad \lim_n f(a_n) = f(C) = \lim_n f(b_n)$$

By the squeeze theorem, since $\forall n : f(a_n) \leq u \leq f(b_n)$

$$(287) \quad f(C) = u$$

Thus it is proved that there exists such a C for $f(A) \leq u \leq f(B)$. If $f(A) \geq u \geq f(B)$, then

$$(288) \quad -f(A) \leq -u \leq -f(B)$$

Let $g(x) = -f(x)$ and $v = -u$, then

$$(289) \quad g(A) \leq v \leq g(B)$$

By the earlier result, $\exists C \in [A, B] : g(C) = v$, then $-f(C) = -u$ or $f(C) = u$. $\square$

Theorem 12.8. *If $f : M \rightarrow N$ is continuous and S is compact in M , then $f[S]$ is compact.*

Corollary 12.9 (Bounded Value Theorem). *If $f : M \rightarrow N$ is continuous and S is compact in M , then $f[S]$ is bounded.*

Theorem 12.10 (Bounded Value Theorem). *If $f : [a, b] \rightarrow \mathbb{R}$ is a continuous function, then f is bounded.*

$$(290) \quad \exists C : \forall x \in [a, b] : m \leq f(x) \leq M$$

Proof. Let $f : [A, B] \rightarrow \mathbb{R}$ be any continuous function. Assume without loss of generality

$$(291) \quad \forall C : \exists x \in [A, B] : f(x) > C$$

Since $\mathbb{Z}^+ \subset \mathbb{R}$

$$(292) \quad \forall n \in \mathbb{Z}^+, \exists x \in [A, B] : f(x) > n$$

Then there is a sequence x such that for all n

$$(293) \quad x_n \in [A, B] \wedge f(x_n) > n$$

By the point peak theorem, there is a monotonic subsequence $x \circ k$. Since x is bounded, the subsequence $x \circ k$ is bounded. Since $x \circ k$ is bounded and monotone

$x \circ k$ converges by the monotone convergence theorem. Since f is continuous, $f(x \circ k)$ also converges. Since $x \circ k$ is a subsequence

$$(294) \quad k_i \geq i$$

By trichotomy, for all i

$$(295) \quad f(x_{k_i}) > i$$

Thus $f(x \circ k)$ diverges, which is a contradiction. Therefore f is bounded above. $\square$

Theorem 12.11 (Extreme Value Theorem). *If $f : [A, B] \rightarrow \mathbb{R}$ is continuous*

$$(296) \quad \exists M \in [A, B], \forall x \in [A, B] : f(x) \leq f(M)$$

Proof. By the boundedness theorem, there is a bound, B to the continuous function. Define four sequences a, b, c, e such that

$$(297) \quad a_1 := f(A)$$

$$(298) \quad b_1 := B$$

For all n :

$$(299) \quad c_n := \frac{a_n + b_n}{2}$$

$$(300) \quad e_n := b_n - a_n$$

If c_n is a bound of the continuous function ($\forall x \in [A, B] : c_n \geq f(x)$):

$$(301) \quad a_{n+1} := a_n$$

$$(302) \quad b_{n+1} := c_n$$

Else

$$(303) \quad a_{n+1} := c_n$$

$$(304) \quad b_{n+1} := b_n$$

If c_n is a bound

$$(305) \quad a_{n+1} = a_n$$

$$(306) \quad b_{n+1} = c_n = \frac{a_n + b_n}{2}$$

$$(307) \quad e_{n+1} = \frac{a_n + b_n}{2} - a_n = \frac{b_n - a_n}{2}$$

If c_n is not a bound

$$(308) \quad a_{n+1} = c_n = \frac{a_n + b_n}{2}$$

$$(309) \quad b_{n+1} = b_n$$

$$(310) \quad e_{n+1} = b_n - \frac{a_n + b_n}{2} = \frac{b_n - a_n}{2}$$

Since $\forall n : a_n < c_n < b_n$, a_n is increasing and b_n is decreasing. Since they are also bounded by (A, B) , by the monotone convergence theorem, they both converge. From the definition of e_n

$$(311) \quad e_n = b_n - a_n$$

So

$$(312) \quad e_{n+1} = \frac{e_n}{2}$$

By the lemma 12.6

$$(313) \quad e_n = \frac{1}{2^{n-1}} e_1 = \frac{1}{2^{n-1}} (B - A)$$

which converges to zero. Therefor

$$(314) \quad \lim_n (b_n - a_n) = 0$$

$$(315) \quad L := \lim_n b_n = \lim_n a_n$$

□

13. CONNECTED SPACES

Definition 13.1 (Connected Space). A set is *connected*, if it has no non-empty, proper, clopen subset. A set that is not connected is *disconnected*.

Theorem 13.2. *If $f : M \rightarrow N$ is continuous, and M is connected, then $f[M]$ is connected.*

Proof. Suppose $f[M]$ is disconnected. Then there is some non-empty, proper, clopen subset A .

$$(316) \quad A \subsetneq f[M]$$

The pre-image of A , then is a non-empty, proper subset of M

$$(317) \quad f^{-1}[A] \subsetneq M$$

By the open and closed set conditions, the preimage $f^{-1}[A]$ is clopen. But by assumption, there is no non-empty, proper, clopen subset. So $f[M]$ must be connected. □

Lemma 13.3. *If I is a connected interval on $\mathbb{R}$,*

$$(318) \quad \forall a, b \in I : a < b \Rightarrow \exists c \in I : a < c < b$$

Proof. Suppose its not,

$$(319) \quad \exists a, b \in I : a < b \wedge c \in (-\infty, a] \cup [b, \infty)$$

Let $J := (-\infty, a] \cup [b, \infty)$. The subsets of J , $(-\infty, a]$ and its conjugate $[b, \infty)$ are both closed, therefor both are clopen, and I is disconnected. □

Theorem 13.4 (Generalized Intermediate Value Theorem). *If $f : M \rightarrow \mathbb{R}$ is continuous and $I \subset M$ is connected, then*

$$(320) \quad \begin{aligned} & \forall a, b \in I : f(a) < f(b) \\ & \Rightarrow \exists c \in I : f(a) < f(c) < f(b) \end{aligned}$$

Proof. Since I is connected, $f[I]$ is connected. Hence there is a $\gamma \in f[I]$

$$(321) \quad f(a) < \gamma < f(b)$$

Therefor there is a $c \in I$ such that

$$(322) \quad \exists c \in I : f(a) < f(c) < f(b)$$

□

14. DIFFERENTIABLE FUNCTIONS

Definition 14.1 (Differentiable Function). A function f is *differentiable* at a if the following exists.

$$(323) \quad f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

This limit $f'(a)$ is called the *derivative* of f at a

Theorem 14.2. *Every differentiable function is continuous.*

Proof.

$$(324) \quad f(x) - f(a) = \frac{f(x) - f(a)}{x - a}(x - a)$$

$$(325) \quad \lim_{x \rightarrow a} [f(x) - f(a)] = \lim_{x \rightarrow a} \left[\frac{f(x) - f(a)}{x - a}(x - a) \right]$$

If $f : [A, B] \rightarrow \mathbb{R}$ is differentiable at $a \in [A, B]$, then $f'(a)$ exists.

$$(326) \quad f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

$$(327) \quad \begin{aligned} \lim_{x \rightarrow a} [f(x) - f(a)] &= f'(a) \cdot 0 \\ &= 0 \end{aligned}$$

$$(328) \quad \begin{aligned} \lim_{x \rightarrow a} f(x) &= \lim_{x \rightarrow a} f(a) \\ \lim_{x \rightarrow a} f(x) &= f(a) \end{aligned}$$

Thus $f(x)$ is continuous at a □

Theorem 14.3 (Chain Rule). *If $f : [a, b] \rightarrow I$ and $g : I \rightarrow \mathbb{R}$ are differentiable, then*

$$(329) \quad (f \circ g)'(x) = (f' \circ g)(x)g'(x)$$

Proof. Define u, v such that

$$(330) \quad u(t) := f' \circ g(x) - \frac{(f \circ g)(x) - (f \circ g)(t)}{g(x) - g(t)}$$

$$(331) \quad v(t) := g'(x) - \frac{g(x) - g(t)}{x - t}$$

The limit of u at x is then

$$(332) \quad \lim_{t \rightarrow x} u(t) = (f' \circ g)(x) - \lim_{t \rightarrow x} \frac{(f \circ g)(x) - (f \circ g)(t)}{g(x) - g(t)}$$

$$(333) \quad = (f' \circ g)(x) - \lim_{g(t) \rightarrow g(x)} \frac{(f \circ g)(x) - (f \circ g)(t)}{g(x) - g(t)}$$

$$(334) \quad = (f' \circ g)(x) - (f' \circ g)(x) = 0$$

And the limit of v at x is also

$$(335) \quad \lim_{t \rightarrow x} v(t) = g'(x) - \lim_{t \rightarrow x} \frac{g(x) - g(t)}{x - t}$$

$$(336) \quad = g'(x) - g'(x) = 0$$

Rearranging the definition of v gives

$$(337) \quad g(x) - g(t) = (x - t) [g'(x) - v(t)]$$

and the definition of u

$$(338) \quad (f \circ g)(x) - (f \circ g)(t) = [g(x) - g(t)] [f' \circ g(x) - u(t)]$$

Substituting gives

$$(339) \quad (f \circ g)(x) - (f \circ g)(t) = (x - t) [g'(x) - v(t)] [f' \circ g(x) - u(t)]$$

The definition of the limit of the composite function gives

$$(340) \quad (f \circ g)'(x) = \lim_{t \rightarrow x} \frac{(f \circ g)(x) - (f \circ g)(t)}{x - t}$$

$$(341) \quad = \lim_{t \rightarrow x} \frac{x - t}{x - t} [g'(x) - v(t)] [f' \circ g(x) - u(t)]$$

$$(342) \quad = (f' \circ g)(x) \cdot g'(x)$$

□

Lemma 14.4. *If f is continuous on $[a, b]$ and differentiable on (a, b) , then*

$$(343) \quad f(a) = f(b) \Rightarrow \exists c \in (a, b) : f'(c) = 0$$

Proof. Since f is continuous on $[a, b]$, f achieves a maximum $M \in [a, b]$ and a minimum $m \in [a, b]$.

Case 1. Suppose m and M are both end points of $[a, b]$. Then

$$(344) \quad f(a) = f(b) \Rightarrow f(m) = f(M)$$

Since $f(m)$ and $f(M)$ are minimum and maximums of f respectively

$$(345) \quad \forall x \in [a, b] : f(m) \leq f(x) \leq f(M)$$

along with $f(m) = f(M)$, implies

$$(346) \quad \forall x \in [a, b] : f(m) = f(x) = f(M)$$

So f is a constant function on $[a, b]$, and the derivative of the constant function is

$$(347) \quad \forall x \in (a, b) : f'(x) = 0$$

Case 2. Now suppose that m or M is not an end point of $[a, b]$. Then there is a $c \in (a, b)$ such that f has a local extrema at c . Since the derivative at a local extrema is zero.

$$(348) \quad f'(c) = 0$$

So for all cases,

$$(349) \quad \exists c \in (a, b) : f'(c) = 0$$

□

Theorem 14.5 (Generalized Mean Value Theorem). *If f and g are continuous on $[a, b]$ and differentiable on (a, b) , then*

$$(350) \quad \exists c \in (a, b) : [f(b) - f(a)] g'(c) = [g(b) - g(a)] f'(c)$$

Proof. Define h such that

$$(351) \quad h(x) := [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x)$$

Since f and g both are continuous on $[a, b]$ and differentiable on (a, b) , h is continuous on $[a, b]$ and differentiable on (a, b) . So the derivative of h on (a, b) is

$$(352) \quad h'(x) = [f(b) - f(a)]g'(x) - [g(b) - g(a)]f'(x)$$

At the end points

$$\begin{aligned} h(a) &= f(b)g(a) - f(a)g(a) - g(b)f(a) + g(a)f(a) \\ &= f(b)g(a) - f(a)g(b) \end{aligned}$$

And

$$\begin{aligned} h(b) &= f(b)g(b) - f(a)g(b) - g(b)f(b) + g(a)f(b) \\ (353) \quad &= f(b)g(a) - f(a)g(b) \end{aligned}$$

So

$$(354) \quad h(a) = h(b)$$

By the above lemma

$$(355) \quad \exists c \in (a, b) : h'(c) = 0$$

So there is a $c \in (a, b)$ such that

$$(356) \quad h'(c) = 0 = [f(b) - f(a)]g'(c) - [g(b) - g(a)]f'(c)$$

or

$$(357) \quad [f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c)$$

□

Theorem 14.6 (L'hôpital's Rule). *If f and g are differentiable functions and*

$$(358) \quad \begin{aligned} \lim_{x \rightarrow a} f(x) &= 0 \\ \lim_{x \rightarrow a} g(x) &= 0 \end{aligned}$$

Then

$$(359) \quad \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

Proof. Choose some ε and ε' such that $\varepsilon > \varepsilon' > 0$. By 359, there is some $\delta > 0$ such that $\forall x \in [a, b]$

$$(360) \quad 0 < |x - c| < \delta \Rightarrow \left| \frac{f'(x)}{g'(x)} - L \right| < \varepsilon'$$

For all $(x, y) \subset (c, c + \delta)$ or $(x, y) \subset (c - \delta, c)$, by generalized mean value theorem, there is a $t \in (x, y)$ such that

$$(361) \quad \frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f'(t)}{g'(t)}$$

Hence

$$\left| \frac{f(x) - f(y)}{g(x) - g(y)} - L \right| < \varepsilon'$$

$$(362) \quad \begin{aligned} L - \varepsilon' &< \frac{f(x) - f(y)}{g(x) - g(y)} < L + \varepsilon' \\ L - \varepsilon' &\leq \lim_{x \rightarrow c} \frac{f(x) - f(y)}{g(x) - g(y)} \leq L + \varepsilon' \end{aligned}$$

Assume that $\lim_{x \rightarrow c} f(x) = 0$ and $\lim_{x \rightarrow c} g(x) = 0$, then

$$(363) \quad L - \varepsilon' \leq \frac{f(y)}{g(y)} \leq L + \varepsilon'$$

$$(364) \quad \left| \frac{f(y)}{g(y)} - L \right| \leq \varepsilon' < \varepsilon$$

So for every $\varepsilon > 0$, there is a $\delta > 0$ such that for all $y \in [a, b]$

$$(365) \quad |y - c| < \delta \Rightarrow \left| \frac{f(y)}{g(y)} - L \right| < \varepsilon$$

Or

$$(366) \quad \lim_{y \rightarrow c} \frac{f(y)}{g(y)} = L$$

□

15. INTEGRABLE FUNCTIONS

Definition 15.1 (Partition). A *partition* on $[a, b]$ is a finite, increasing sequence $x : [0, n] \rightarrow [a, b]$, such that

$$(367) \quad \begin{aligned} x_0 &= a \\ x_n &= b \end{aligned}$$

The sequence of lengths of partitions Δx_i is defined simply as

$$(368) \quad \Delta x_i := x_{i-1} - x_i$$

The *mesh* of a partition is given by

$$(369) \quad \text{mesh } x := \max_i \Delta x_i$$

A sequence of sample points $t : [0, n] \rightarrow [a, b]$ for a partition

$$(370) \quad \forall i : x_{i-1} \leq t_i \leq x_i$$

A *partition pair* is a tuple of sequences, the partition x and one of its sequence of sample points t .

Definition 15.2 (Partition Sum). The *partition sum* of a function on f with partition pair x, t is given by

$$(371) \quad R(f, x, t) := \sum_{i=1}^n f(t_i) \Delta x_i$$

Definition 15.3 (Integral). A function is *integrable* on $[a, b]$ if there is an $I \in \mathbb{R}$ such that, for all $\varepsilon > 0$ there is a $\delta > 0$ such that for all partitions x

$$(372) \quad \text{mesh } x < \delta \Rightarrow |R(f, x, t) - I| < \varepsilon$$

If f is integrable on $[a, b]$, I is the integral of f over $[a, b]$ and it is denoted by

$$(373) \quad \int_a^b f(x) dx = I = \lim_{\text{mesh } x \rightarrow 0} \sum_{i=1}^n f(t_i) \Delta x_i$$

Definition 15.4 (Upper Sum).

$$(374) \quad U(f, x) := \sum_{i=1}^n M_i \Delta x_i$$

where

$$(375) \quad M_i := \sup\{f(x) : t_{i-1} \leq x \leq t_i\}$$

Definition 15.5 (Lower Sum).

$$(376) \quad L(f, x) := \sum_{i=1}^n m_i \Delta x_i$$

where

$$(377) \quad m_i := \inf\{f(x) : t_{i-1} \leq x \leq t_i\}$$

Lemma 15.6. *If t^* is a refinement of t , such that $t = t^* \circ p$*

$$(378) \quad \sum_{j=p_{i-1}+1}^{p_i} a(t_j^* - t_{j-1}^*) = a(t_i - t_{i-1})$$

Proof.

$$(379) \quad \sum_{j=p_{i-1}+1}^{p_i} a(t_j^* - t_{j-1}^*) = a \sum_{j=p_{i-1}+1}^{p_i} (t_j^* - t_{j-1}^*)$$

$$(380) \quad = a \sum_{j=i-1+1}^i (t_{p_j}^* - t_{p_{j-1}}^*)$$

$$(381) \quad = a \sum_{j=i}^i (t_j - t_{j-1})$$

$$(382) \quad = a(t_i - t_{i-1})$$

□

Theorem 15.7. *If x^* is a refinement of x , then*

$$(383) \quad \begin{aligned} L(f, x) &\leq L(f, x^*) \\ U(f, x) &\geq U(f, x^*) \end{aligned}$$

Proof. Define M , M^* , m and m^* such that

$$\begin{aligned} M_i &= \sup\{f(x) : t_{i-1} \leq x \leq t_i\} \\ M_i^* &= \sup\{f(x) : t_{i-1}^* \leq x \leq t_i^*\} \\ m_i &= \inf\{f(x) : t_{i-1} \leq x \leq t_i\} \\ m_i^* &= \inf\{f(x) : t_{i-1}^* \leq x \leq t_i^*\} \end{aligned}$$

Let $p : [0, n] \rightarrow [0, n^*]$ be a finite sequence such that

$$(384) \quad t = t^* \circ p$$

Then M and m can be written as

$$\begin{aligned} M_i &= \sup\{f(x) : t_{p_{i-1}}^* \leq x \leq t_{p_i}^*\} \\ m_i &= \inf\{f(x) : t_{p_{i-1}}^* \leq x \leq t_{p_i}^*\} \end{aligned}$$

Thus $\forall i \in [0, n] : \forall j \in [p_{i-1}, p_i] : m_j^* \geq m_i$ and $M_j^* \leq M_i$, thus

$$(385) \quad \sum_{j=p_{i-1}+1}^{p_i} m_j^*(t_j^* - t_{j-1}^*) \geq \sum_{j=p_{i-1}+1}^{p_i} m_i(t_j^* - t_{j-1}^*)$$

$$(386) \quad \sum_{j=p_{i-1}+1}^{p_i} M_j^*(t_j^* - t_{j-1}^*) \leq \sum_{j=p_{i-1}+1}^{p_i} M_i(t_j^* - t_{j-1}^*)$$

By the above lemma

$$(387) \quad \sum_{j=p_{i-1}+1}^{p_i} m_j^*(t_j^* - t_{j-1}^*) \geq m_i(t_i - t_{i-1})$$

$$(388) \quad \sum_{j=p_{i-1}+1}^{p_i} M_j^*(t_j^* - t_{j-1}^*) \leq M_i(t_i - t_{i-1})$$

And

$$(389) \quad \sum_{i=1}^n \sum_{j=p_{i-1}+1}^{p_i} m_j^*(t_j^* - t_{j-1}^*) \geq \sum_{i=1}^n m_i(t_i - t_{i-1})$$

$$(390) \quad \sum_{i=1}^n \sum_{j=p_{i-1}+1}^{p_i} M_j^*(t_j^* - t_{j-1}^*) \leq \sum_{i=1}^n M_i(t_i - t_{i-1})$$

And finally

$$(391) \quad \sum_{i=1}^{n^*=p_n} m_i^*(t_i^* - t_{i-1}^*) \geq \sum_{i=1}^n m_i(t_i - t_{i-1})$$

$$(392) \quad \sum_{i=1}^{n^*=p_n} M_i^*(t_i^* - t_{i-1}^*) \leq \sum_{i=1}^n M_i(t_i - t_{i-1})$$

Giving

$$(393) \quad L(f, t^*) \geq L(f, t)$$

$$(394) \quad U(f, t^*) \leq U(f, t)$$

□

Lemma 15.8. *For a partition P*

$$(395) \quad L(f, P) \leq U(f, P)$$

Proof. Define M and m such that

$$(396) \quad M_i = \sup\{f(x) : t_{i-1} \leq x \leq t_i\}$$

$$(397) \quad m_i = \inf\{f(x) : t_{i-1} \leq x \leq t_i\}$$

then for every $i : m_i \leq M_i$, thus

$$(398) \quad \sum_{i=1}^n m_i(t_i - t_{i-1}) \leq \sum_{i=1}^n M_i(t_i - t_{i-1})$$

Hence $L(f, t) \leq U(f, t)$ □

Theorem 15.9. For all partitions, P_1, P_2 ,

$$(399) \quad L(f, P_1) \leq U(f, P_2)$$

Proof. Let t^* be the common refinement of t_1 and t_2 , then

$$(400) \quad L(f, t_1) \leq L(f, t^*)$$

$$(401) \quad U(f, t^*) \leq U(f, t_2)$$

By the above lemma

$$(402) \quad L(f, t^*) \leq U(f, t^*)$$

Hence

$$(403) \quad L(f, t_1) \leq U(f, t_2)$$

□

Theorem 15.10. A bounded function f is integrable on $[a, b]$, if for every $\varepsilon > 0$, there is a partition x such that

$$(404) \quad U(f, x) - L(f, x) < \varepsilon$$

Proof. Let $\varepsilon > 0$ and suppose such a partition x exists such that

$$(405) \quad U(f, x) - L(f, x) < \varepsilon$$

□

16. PRIMITIVES

Definition 16.1 (Primitive). Let f be continuous on (a, b) . The function F is a primitive of f , if F is differentiable on (a, b) and

$$(406) \quad \forall x \in (a, b) : F'(x) = f(x)$$

Lemma 16.2. If f is continuous on $[a, b]$, then the function $F(x) = \int_a^x f(t) dt$ is uniformly continuous on $[a, b]$.

Lemma 16.3. If f is continuous on $[a, b]$, then the function $F(x) = \int_a^x f(t) dt$ is differentiable on (a, b) .

Theorem 16.4. If f is continuous on $[a, b]$, then the function $F(x) = \int_a^x f(t) dt$ is a primitive of f .

Theorem 16.5. If f is integrable on $[a, b]$, F is differentiable on $[a, b]$, and F is a primitive of f , then

$$(407) \quad \int_a^b f(x) dx = F(b) - F(a)$$

APPENDIX A. REAL NUMBERS

A.1. Rational Sequences.

Definition A.1 (Sequence). A *sequence* of rationals is a function whose domain is the positive integers and co-domain is a subset of rationals.

$$(408) \quad \begin{aligned} s : \mathbb{Z}^+ &\rightarrow \mathbb{Q} \\ n &\mapsto s_n \end{aligned}$$

Let $\mathcal{S}$ denote the set of all rational sequences.

Definition A.2 (Constant Sequence). A *constant sequence* is a sequence $c^* \in \mathcal{S}$ in which all the values are the same.

$$(409) \quad \forall n \in \mathbb{Z}^+ : c_n^* = c$$

where $c \in \mathbb{Q}$. Let $\mathbb{Q}^*$ denote the set of all constant, rational sequences.

Definition A.3 (Convergent Sequence). A *convergent sequence* of rationals is a sequence $s \in \mathcal{S}$, in which there is an $L \in \mathbb{Q}$ such that

$$(410) \quad \forall \varepsilon \in \mathbb{Q}^+ : \exists N \in \mathbb{Z}^+ : n > N \Rightarrow |s_n - L| < \varepsilon$$

If s converges to L , then L is called the *limit* of the sequence, denoted by

$$(411) \quad \lim_n s = L$$

Let $\mathcal{C}$ denote the set of all convergent, rational sequences.

Theorem A.4. *Every constant sequence is convergent, the limit being the value of the function.*

$$(412) \quad \mathbb{Q}^* \subset \mathcal{C}$$

and for $c \in \mathbb{Q}$

$$(413) \quad c^* : n \mapsto c \Rightarrow \lim_n c_n^* = c$$

Proof. Let $c^* : n \mapsto c$ be a rational, constant sequence. Choose any $\varepsilon \in \mathbb{Q}^+$ and any $N \in \mathbb{Z}^+$. For all n , including $n > N$,

$$(414) \quad |c_n^* - c| = 0 < \varepsilon$$

which is the criterion for a convergent sequence

$$(415) \quad \lim_n c_n^* = c$$

□

Definition A.5 (Fundamental Sequence). A *fundamental sequence* of rationals is a sequence $s \in \mathcal{S}$, where

$$(416) \quad \forall \varepsilon \in \mathbb{Q}^+ : \exists N \in \mathbb{Z}^+ : m, n > N \Rightarrow |s_m - s_n| < \varepsilon$$

Let $\mathcal{F}$ denote the set of all fundamental, convergent sequences.

Theorem A.6. *Every convergent sequence is fundamental.*

$$(417) \quad \mathcal{C} \subset \mathcal{F}$$

Proof. Let s be a convergent sequence $s \in \mathcal{F}$. Choose $\varepsilon \in \mathbb{Q}^+$, which implies $\exists N \in \mathbb{Z}^+$, such that for $m, n > N$, the convergent criterion holds.

$$(418) \quad |s_m - s_n| = |(s_m - L) - (s_n - L)|$$

By triangle inequality

$$(419) \quad |(s_m - L) - (s_n - L)| \leq |s_m - L| + |s_n - L|$$

Define $\varepsilon' := 2\varepsilon$. By the definition of a convergent sequence, for all $n > N$

$$(420) \quad |s_n - L| < \varepsilon = \frac{\varepsilon'}{2}$$

By $a < b \wedge c < d \Rightarrow a_c < b + d$

$$(421) \quad |s_m - L| + |s_n - L| < \frac{\varepsilon'}{2} + \frac{\varepsilon'}{2} = \varepsilon'$$

Therefor

$$(422) \quad |s_m - s_n| < \varepsilon'$$

which is the fundamental criterion. $\square$

Definition A.7 (Bounded Sequence). A sequence $s \in \mathcal{S}$ is *completely bounded* by the interval (b, B) if

$$(423) \quad \forall n \in \mathbb{Z}^+ : b < s_n < B$$

Let $\mathcal{B}$ denote the set of all bounded sequences

Theorem A.8. *Every fundamental sequence is bounded.*

$$(424) \quad \mathcal{F} \subset \mathcal{B}$$

Proof. Choose a fundamental sequence $s \in \mathcal{F}$ and choose a $\varepsilon \in \mathbb{Q}^+$. This implies the existence of $N \in \mathbb{Z}^+$ where if $m, n > N$ the fundamental criterion holds. Define $M := N + 1$. Since $\{|s_n| : n \leq M\}$ is a finite set, $\max\{|s_n| : n \leq M\}$ is just a rational number. Define a number

$$(425) \quad B := \max\{|s_n| : n \leq M\} + \varepsilon$$

The number $B - \varepsilon$ bounds the sequence for $n \leq M$; for $n > M$:

$$|s_n| = |s_n - s_M + s_M|$$

By the triangle inequality

$$|s_n| \leq |s_n - s_M| + |s_M|$$

From the definition of a fundamental sequence

$$(426) \quad |s_n - s_M| < \varepsilon$$

Since $B - \varepsilon$ is a bound for $n \leq M$

$$(427) \quad |s_M| < B - \varepsilon$$

By $a < b \wedge c < d \Rightarrow a + c < b + d$

$$(428) \quad |s_n| < \varepsilon + B - \varepsilon = B$$

$\square$

A.2. The Sequence Subrings.

Definition A.9. Define addition of two sequences as

$$(429) \quad s + t := i \mapsto s_i + t_i$$

Subtraction of two sequences by

$$(430) \quad s - t := i \mapsto s_i - t_i$$

and multiplication of two sequences as

$$(431) \quad st := i \mapsto s_i t_i$$

Theorem A.10. *The sum of two fundamental sequences is fundamental.*

Proof. By the triangle inequality

$$(432) \quad |s_m - s_n + t_m - t_n| \leq |s_m - s_n| + |t_m - t_n|$$

By the definition of fundamental sequences, $\forall \varepsilon_s, \varepsilon_t \in \mathbb{Q}^+$, there is a $N_s, N_t \in \mathbb{Z}^+$ and $N := \max\{N_s, N_t\}$, such that if $m, n > N$

$$(433) \quad \begin{aligned} |s_m - s_n| &< \varepsilon_s \\ |t_m - t_n| &< \varepsilon_t \end{aligned}$$

By $a < b \wedge c < d \Rightarrow a + c < b + d$

$$(434) \quad |s_m - s_n| + |t_m - t_n| < \varepsilon_s + \varepsilon_t$$

Let $\varepsilon := \varepsilon_s + \varepsilon_t$

$$(435) \quad |s_m - s_n + t_m - t_n| < \varepsilon$$

$$(436) \quad |(s_m + t_m) - (s_n + t_n)| < \varepsilon$$

□

Theorem A.11. *The product of two fundamental sequences is fundamental.*

Proof.

$$(437) \quad |s_m t_m - s_n t_n| = |s_m t_m - s_m t_n + s_m t_n - s_n t_n|$$

$$(438) \quad = |s_m(t_m - t_n) + t_n(s_m - s_n)|$$

By the triangle identity

$$(439) \quad |s_m(t_m - t_n) + t_n(s_m - s_n)| \leq |s_m(t_m - t_n)| + |t_n(s_m - s_n)|$$

Let b_s and b_t be bounds of sequences s and t respectively. Define $B := 2 \max\{b_s, b_t\}$, then by the definition of bounding and order transitivity

$$(440) \quad |s_m| < \frac{B}{2} \wedge |t_n| < \frac{B}{2}$$

The definition of a closing sequences holds for $\frac{\varepsilon}{B}$, so

$$(441) \quad |s_m - s_n| < \frac{\varepsilon}{B} \wedge |t_m - t_n| < \frac{\varepsilon}{B}$$

By $0 < a < b \wedge 0 < c < d \Rightarrow ac < bd$

$$(442) \quad \begin{aligned} |s_m| |t_m - t_n| &< \frac{B}{2} \frac{\varepsilon}{B} = \frac{\varepsilon}{2} \\ |t_n| |s_m - s_n| &< \frac{B}{2} \frac{\varepsilon}{B} = \frac{\varepsilon}{2} \end{aligned}$$

By $a < b \wedge c < d \Rightarrow a + c < b + d$

$$(443) \quad |s_m||t_m - t_n| + |t_n||s_m - s_n| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

Therefor

$$(444) \quad |s_m t_m - s_n t_n| < \varepsilon$$

□

Theorem A.12. *The sum of two convergent sequences is convergent*

$$(445) \quad \lim_n s_n + \lim_n t_n = \lim_n (s_n + t_n)$$

Theorem A.13. *The product of two convergent sequences is convergent*

$$(446) \quad \lim_n s_n \cdot \lim_n t_n = \lim_n (s_n t_n)$$

A.3. Ordered Rings.

Definition A.14.

$$(447) \quad \mathcal{Z} := \left\{ \{s_n\} \in \mathcal{C} : \lim_n s_n = 0 \right\}$$

Definition A.15 (Positive). The set of *positive* fundamental sequences are sequences that have a tail that eventually is greater than some positive rational.

$$(448) \quad \mathcal{F}^+ := \left\{ \{s_n\} \in \mathcal{F} \mid \exists \varepsilon \in \mathbb{Q}^+, N \in \mathbb{Z}^+ : n > N \Rightarrow s_n > \varepsilon \right\}$$

Theorem A.16. *For all fundamental sequences $\{s_n\} \in \mathcal{F}$, exactly one of the following is true: $\{s_n\} \in \mathcal{Z}$, $\{s_n\} \in \mathcal{F}^+$, or $\{-s_n\} \in \mathcal{F}^+$*

Proof. If a sequence $\{s_n\}$ is fundamental then,

$$(449) \quad \forall \varepsilon \in \mathbb{Q}^+, \exists N \in \mathbb{Z}^+, \forall m, n > N : |s_m - s_n| < \varepsilon$$

but not $\mathcal{Z}$

$$(450) \quad \exists \varepsilon' \in \mathbb{Q}^+, \forall N \in \mathbb{Z}^+, \exists n > N : |s_n| \geq \varepsilon'$$

If there is such an ε' , then there must be an r such that $\varepsilon' > r > 0$

$$(451) \quad \forall N \in \mathbb{Z}^+, \exists n > N : |s_n| > r$$

Choose some ε , such that $r > \varepsilon > 0$ and N

$$(452) \quad \forall m, n > N \Rightarrow |s_m - s_n| < \varepsilon$$

Choose $n > N$ such that

$$(453) \quad |s_n| > r$$

Use the definition of a fundamental sequence

$$(454) \quad s_n - \varepsilon < s_m < s_n + \varepsilon$$

If $s_n > 0$, then $s_n > r$

$$(455) \quad r - \varepsilon < s_n - \varepsilon < s_m$$

Since $r > \varepsilon$

$$(456) \quad 0 < r - \varepsilon < s_m$$

So that, for every $m > N$

$$(457) \quad 0 < s_m$$

If $s_n < 0$, then $s_n < -r$

$$(458) \quad s_m < \varepsilon + s_n < \varepsilon - r$$

Since $r > \varepsilon$

$$(459) \quad s_m < \varepsilon - r < 0$$

So that, for every $m > N$

$$(460) \quad s_m < 0$$

Therefor, for every sequence $\{s_n\} \in \mathbb{F} - \mathbb{Z}$, one of the following is true

$$(461) \quad \exists N : n > N \Rightarrow 0 < s_n$$

$$(462) \quad \exists N : n > N \Rightarrow s_n < 0$$

The first one implies that $\{s_n\} \in \mathcal{F}^+$, for the second one, take $\{-s_n\}$, then

$$(463) \quad \exists N : n > N \Rightarrow 0 < -s_n$$

so that $\{-s_n\} \in \mathcal{F}^+$ □

Theorem A.17 (Closure). *Addition and multiplication are closed on $\mathcal{F}^+$*

Proof. Choose two sequences $s, t \in \mathcal{F}^+$, then there is an N_s, N_t such that

$$(464) \quad n > N_s \Rightarrow s_n > 0$$

$$(465) \quad n > N_t \Rightarrow t_n > 0$$

Define $N := \max\{N_s, N_t\}$, then for any $n > N$

$$(466) \quad s_n > 0 \wedge t_n > 0$$

Since addition and multiplication are closed on positive rationals

$$(467) \quad s_n + t_n > 0 \wedge s_n t_n > 0$$

And

$$(468) \quad s + t \in \mathcal{F}^+$$

$$(469) \quad s \cdot t \in \mathcal{F}^+$$

□

Definition A.18.

$$(470) \quad x > y \iff x - y \in \mathcal{F}$$

$$(471) \quad x < y \iff y - x \in \mathcal{F}$$

A.4. The Real Field.

Definition.

$$(472) \quad \mathcal{Z} := \left\{ \{s_n\} \in \mathcal{C} : \lim_n s_n = 0 \right\}$$

Theorem A.19. $\mathcal{Z}$ is a subgroup of $(\mathcal{S}, +)$

Proof. The additive identity 0^* is in $\mathcal{Z}$, since it converges to zero.

$$(473) \quad \lim_n 0^* = 0$$

And addition and subtraction is closed on $\mathcal{Z}$. If $a, b \in \mathcal{Z}$

$$(474) \quad \lim_n (a_n + b_n) = \lim_n a_n + \lim_n b_n = 0 + 0 = 0$$

$$(475) \quad \lim_n (a_n - b_n) = \lim_n a_n - \lim_n b_n = 0 - 0 = 0$$

□

Lemma A.20. *The product of a bounded sequence and a sequence in $\mathcal{Z}$ is in $\mathcal{Z}$*

$$(476) \quad \forall s \in \mathcal{B}, t \in \mathcal{Z} : st \in \mathcal{Z}$$

Proof. Since s is bounded

$$(477) \quad \forall n \in \mathbb{Z}^+ : b < s_n < B$$

$$(478) \quad |s_n| < |B|$$

Fix $\varepsilon \in \mathbb{Q}^+$ and $N \in \mathbb{Z}^+$, then for all $n > N$

$$(479) \quad |t_n| < \varepsilon$$

Then for all $n > N$

$$(480) \quad |s_n||t_n| < |B|\varepsilon|s_n \cdot t_n| < |B|\varepsilon$$

Let $\varepsilon' := |B|\varepsilon$, then for all $\varepsilon' \in \mathbb{Q}^+$

$$(481) \quad n > N \Rightarrow |s_n t_n| < \varepsilon'$$

or

$$(482) \quad \lim_n s_n t_n = 0$$

and

$$(483) \quad s \cdot t \in \mathcal{Z}$$

□

Definition A.21 (Equivalence). Two sequences are equivalent mod $\mathcal{Z}$ if and only if their difference is in $\mathcal{Z}$

$$(484) \quad s \equiv_{\mathcal{Z}} t \iff s - t \in \mathcal{Z}$$

Theorem A.22 (Equivalence Reflexivity).

$$(485) \quad a \equiv_{\mathcal{Z}} a$$

Proof.

$$(486) \quad a - a : n \mapsto a_n - a_n = 0$$

$$(487) \quad a - a : n \mapsto 0$$

$$(488) \quad a - a = 0^* \in \mathcal{Z}$$

$$(489) \quad a \equiv_{\mathcal{Z}} a$$

□

Theorem A.23 (Equivalence Symmetry).

$$(490) \quad a \equiv_{\mathcal{Z}} b \iff b \equiv_{\mathcal{Z}} a$$

Proof.

$$(491) \quad a \equiv_{\mathcal{Z}} b$$

$$(492) \quad a - b \in \mathcal{Z}$$

$$(493) \quad -(a - b) \in \mathcal{Z}$$

$$(494) \quad b - a \in \mathcal{Z}$$

$$(495) \quad b \equiv_{\mathcal{Z}} a$$

□

Theorem A.24 (Equivalence Transitivity).

$$(496) \quad a \equiv_{\mathcal{Z}} b \wedge b \equiv_{\mathcal{Z}} c \Rightarrow a \equiv_{\mathcal{Z}} c$$

Proof.

$$(497) \quad a \equiv_{\mathcal{Z}} b \wedge b \equiv_{\mathcal{Z}} c$$

$$(498) \quad a - b \in \mathcal{Z} \wedge b - c \in \mathcal{Z}$$

Since addition is closed on $\mathbb{Z}$

$$(499) \quad (a - b) + (b - c) \in \mathcal{Z}$$

$$(500) \quad a - c \in \mathcal{Z}$$

$$(501) \quad a \equiv_{\mathcal{Z}} c$$

□

Theorem A.25. *Addition is compatible with equivalence*

$$(502) \quad s \equiv_{\mathcal{Z}} s' \wedge t \equiv_{\mathcal{Z}} t' \Rightarrow s + t \equiv_{\mathcal{Z}} s' + t'$$

Proof. Since $s \equiv_{\mathcal{Z}} s'$ and $t \equiv_{\mathcal{Z}} t'$

$$(503) \quad s - s' \in \mathcal{Z}$$

$$(504) \quad t - t' \in \mathcal{Z}$$

Since $\mathcal{Z}$ is closed on addition

$$(505) \quad (s - s') + (t - t') \in \mathcal{Z}$$

$$(506) \quad (s + t) - (s' + t') \in \mathcal{Z}$$

which implies that

$$(507) \quad s + t \equiv_{\mathcal{Z}} s' + t'$$

□

Theorem A.26. *Multiplication is compatible with equivalence for bounded sequences*

$$(508) \quad \forall s, s', t, t' \in \mathcal{B} : s \equiv_{\mathcal{Z}} s' \wedge t \equiv_{\mathcal{Z}} t' \Rightarrow st \equiv_{\mathcal{Z}} s't'$$

Proof. Assume $s \equiv_{\mathcal{Z}} s'$ and $t \equiv_{\mathcal{Z}} t'$, then

$$(509) \quad s - s' \in \mathcal{Z}$$

$$(510) \quad t - t' \in \mathcal{Z}$$

Since $s \in \mathcal{B}$ and $t' \in \mathcal{B}$

$$(511) \quad s(t - t') \in \mathcal{Z}$$

$$(512) \quad st - st' \in \mathcal{Z}$$

and

$$(513) \quad (s - s')t' \in \mathcal{Z}$$

$$(514) \quad st' - s't' \in \mathcal{Z}$$

Since $\mathcal{Z}$ is closed on addition

$$(515) \quad st - st' + st' - s't' \in \mathcal{Z}$$

$$(516) \quad st - s't' \in \mathcal{Z}$$

By the definition of equivalence

$$(517) \quad st \equiv_{\mathcal{Z}} s't'$$

□

Theorem A.27. *If $x \equiv_{\dagger} y$ and $\lim_n x_n = L$, then $\lim_n y_n = L$*

Proof. Fix $\varepsilon \in \mathbb{Q}^+$

$$(518) \quad \exists N_1 \in \mathbb{Z}^+ : \forall n > N_1 : |x_n - y_n| < \varepsilon$$

$$(519) \quad \exists N_2 \in \mathbb{Z}^+ : \forall n > N_2 : |x_n - L| < \varepsilon$$

Let $N := \max\{N_1, N_2\}$, if $n > N$

$$(520) \quad |x_n - y_n| < \varepsilon$$

$$(521) \quad |x_n - L| < \varepsilon$$

By the triangle identity

$$(522) \quad |y_n - L| = |y_n - x_n + x_n - L| \leq |x_n - y_n| + |x_n - L| < \varepsilon + \varepsilon = 2\varepsilon$$

Let $\varepsilon' := 2\varepsilon$. For all $n > N$

$$(523) \quad |y_n - L| < \varepsilon$$

Or

$$(524) \quad \lim_n y_n = L$$

□

Theorem A.28. *Positive fundamental sequences are closed over equivalence*

$$(525) \quad \forall x \in \mathcal{F}^+ : x \equiv_{\mathcal{Z}} y \Rightarrow y \in \mathcal{F}^+$$

Proof. Since x is in $\mathcal{F}^+$, there is an $r \in \mathcal{Q}^+$ and $N \in \mathbb{Z}^+$ such that for all $n > N$

$$(526) \quad x_n > r > 0$$

Since the rationals are dense, there is a rational s such that

$$(527) \quad r > s > 0$$

$$(528) \quad r - s > 0$$

Since $x - y \in \mathcal{Z}$, there is an $N' \in \mathbb{Z}^+$ such that

$$(529) \quad n > N' > N \Rightarrow |x_n - y_n| < r - s$$

If $n > N'$

$$(530) \quad s - r < y_n - x_n$$

Along with $r < x_n$

$$(531) \quad s - r + r = s < y_n - x_n + x_n = y_n$$

$$(532) \quad \forall n > N' : s > y_n$$

So $y \in \mathcal{F}^+$ □

Definition A.29 (The Real Field). The *real field* is the partition of equivalence classes in $\mathcal{F}$

$$(533) \quad \mathbb{R} := \mathcal{F} / \equiv_{\mathcal{Z}}$$

Theorem A.30. Every fundamental sequence not in $\mathcal{Z}$, has a multiplicative inverse.

$$(534) \quad \forall x \in \mathcal{F} - \mathcal{Z}, \exists x^{-1} \in \mathcal{F} : xx^{-1} \equiv_{\mathcal{Z}} 1^*$$

Proof. If $x \in \mathcal{F} - \mathcal{Z}$, then either $x \in \mathcal{F}^+$ or $-x \in \mathcal{F}^+$. First assume $x \in \mathcal{F}^+$, then there is an $r \in \mathbb{Q}^+$ such that there is a N -tail such that

$$(535) \quad x_n > r > 0$$

Define x^{-1} , such that

$$(536) \quad n > N : x_n^{-1} := \frac{1}{x_n}$$

$$(537) \quad n \geq N : x_n^{-1} := 0$$

And let $y_n := x_{N+n}$ and $y_n^{-1} := x_{N+n}^{-1}$

$$(538) \quad \forall n : y_n y_n^{-1} = x_{N+n} \frac{1}{x_{N+n}} = 1$$

$$(539) \quad yy^{-1} = 1^*$$

Since yy^{-1} is a tail of xx^{-1}

$$(540) \quad xx^{-1} \equiv_{\mathcal{Z}} yy^{-1} = 1^*$$

Now assume $-x \in \mathcal{F}^+$, then there is a $(-x)^{-1} \in \mathcal{F}$

$$(541) \quad (-x)(-x)^{-1} \equiv_{\mathcal{Z}} 1^*$$

$$(542) \quad x[-(-x)^{-1}] \equiv_{\mathcal{Z}} 1^*$$

Therefor each fundamental sequence not in $\mathcal{Z}$ has a multiplicative inverse. □

A.5. Completion of the Field.

Definition A.31 (Fundamental Sequence). A real, fundamental sequence is defined by

$$(543) \quad \forall \varepsilon \in \mathbb{R}^+ : \exists N \in \mathbb{Z}^+ : n, m > N \Rightarrow |x_m - x_n| < \varepsilon$$

Definition A.32 (Convergent Sequence). A real, convergent sequence converges to $L \in \mathbb{R}$ if

$$(544) \quad \forall \varepsilon \in \mathbb{R}^+ : \exists N \in \mathbb{Z}^+ : n > N \Rightarrow |x_n - L| < \varepsilon$$

If L exists it is called the *limit* of the function

$$(545) \quad L = \lim_n x_n$$

Theorem A.33. For every real number there is a greater positive integer.

$$(546) \quad \forall x \in \mathbb{R} : \exists n \in \mathbb{Z}^+ : n^* > x$$

where $n^* := m \mapsto n$ and $\mathbb{Z}^+ \subset \mathbb{Q}$

Proof. Let $x \in \mathbb{R}$ be any real number. By definition it is a fundamental sequence of rationals $x \in \mathcal{C}$. Every fundamental sequence is bounded, so there exists $b, B \in \mathbb{Q}$, such that, for all $i \in \mathbb{Z}^+$

$$(547) \quad b < x_i < B$$

By the Archimedean property of rationals, there exists an $n \in \mathbb{Z}^+$, such that $n > B$.

$$(548) \quad x_i < n$$

Consider the constant sequence $n^* : i \mapsto n$

$$(549) \quad x_i < n = n_i^*$$

Choose any $\varepsilon \in \mathbb{Q}^+$ and any $N \in \mathbb{Z}^+$

$$(550) \quad i > N \Rightarrow x_i < n_i^*$$

which by definition of order on the reals

$$(551) \quad x < n^*$$

□

Theorem A.34. $\mathbb{Q}^*$ is a dense subset of $\mathbb{R}$

$$(552) \quad \forall x \in \mathbb{R}, \varepsilon \in \mathbb{Q}^+ : \exists r \in \mathbb{Q}^* : |x - r| < \varepsilon$$

Proof. Fix $\varepsilon \in \mathbb{Q}^+$ and $x \in \mathbb{R}$. Since x is a rational, fundamental sequence, there is an $N \in \mathbb{Z}^+$ such that

$$(553) \quad m, n > N \Rightarrow |x_m - x_n| < \varepsilon$$

Choose $l > m$, let $r := x_l$

$$(554) \quad n > N \Rightarrow |x_n - r_n| < \varepsilon$$

By the definition of order on $\mathbb{R}$ and the definition of a limit

$$(555) \quad \lim_n |x_n - r_n| < \varepsilon$$

By the definition of the absolute value on $\mathbb{R}$

$$(556) \quad |x - r| < \varepsilon$$

□

Theorem A.35. *Every fundamental sequence of rational, constant sequences converges to a fundamental sequence.*

$$(557) \quad \lim_n r_n^* = r$$

where $r^* : n \mapsto (m \mapsto r_n)$

Proof. Let $r \in \mathcal{F}$ be a rational, fundamental sequence. Choose some $\varepsilon \in \mathbb{R}^+$, then there is some $N \in \mathbb{Z}^+$ such that

$$(558) \quad m, n > N \Rightarrow |r_m - r_n| < \varepsilon$$

Fixing $m > N$, the definition of a limit gives

$$(559) \quad \lim_n r_n = r_m$$

$$(560) \quad \left| \lim_n r_n - r_m \right| = 0 < \varepsilon$$

Define $r^* : m \mapsto (n \mapsto r_m)$, then the limit of the constant sequences r_m^* are the constant

$$(561) \quad \lim_n r_{mn}^* = r_m$$

Giving

$$(562) \quad \left| \lim_n r_n - \lim_n r_{mn}^* \right| < \varepsilon$$

By the definition of order on $\mathbb{R}$

$$(563) \quad |r_n^* - r| < \varepsilon$$

Therefor

$$(564) \quad \forall \varepsilon \in \mathbb{Q}^+ : \exists N \in \mathbb{Z}^+ : n > N \Rightarrow |r_n^* - r| < \varepsilon$$

or more succinctly

$$(565) \quad \lim_n r_n^* = r$$

□

Theorem A.36 (Completeness). *Every real, fundamental sequence converges to a real number.*

Proof. Let x be a real, fundamental sequence. Define a sequence of constant, rational sequences $r^* : \mathbb{Z}^+ \rightarrow \mathbb{Q}^*$ such that

$$(566) \quad x_n - \frac{1}{n} < r_n^* < x_n + \frac{1}{n}$$

or

$$(567) \quad |r_n^* - x_n| < \frac{1}{n}$$

For any $m, n \in \mathbb{Z}^+$, by triangle identity

$$(568) \quad |r_m^* - r_n^*| = |r_m^* - x_m + x_m - x_n + x_n - r_n^*|$$

$$(569) \quad \leq |r_m^* - x_m| + |x_m - x_n| + |r_n^* - x_n|$$

$$(570) \quad < |x_m - x_n| + \frac{1}{m} + \frac{1}{n}$$

Since x is a fundamental sequence, for every $\varepsilon \in \mathbb{R}^+$ there is an $N \in \mathbb{Z}^+$, such that $m, n > N$

$$(571) \quad |x_m - x_n| < \frac{\varepsilon}{3}$$

Choose $N' \in \mathbb{Z}^+ : N' > \frac{3}{\varepsilon}$, and define $M := \max\{N, N'\}$. For all $m, n > M$

$$(572) \quad \begin{aligned} m &> M > \frac{3}{\varepsilon} > 0 \\ n &> M > \frac{3}{\varepsilon} > 0 \end{aligned}$$

implies

$$(573) \quad \begin{aligned} \frac{1}{m} &< \frac{\varepsilon}{3} \\ \frac{1}{n} &< \frac{\varepsilon}{3} \end{aligned}$$

Applying this to the earlier inequality

$$(574) \quad |r_m^* - r_n^*| < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$$

Hence r^* is a fundamental sequence. Therefore by the lemma above

$$(575) \quad \lim_n r_n^* = r$$

where $r^* : n \mapsto (m \mapsto r_n)$. From the definition of r^* , for every $n \in \mathbb{Z}^+$

$$(576) \quad |r_n^* - x_n| < \frac{1}{n}$$

Since $\mathbb{R}$ is unbounded

$$(577) \quad \forall \varepsilon \in \mathbb{R}^+ : \exists N \in \mathbb{Z}^+ : \frac{1}{N} < \varepsilon$$

If $n > N$, then $\frac{1}{n} < \frac{1}{N} < \varepsilon$

$$(578) \quad |r_n^* - x_n| < \frac{1}{N} < \varepsilon$$

Then, for every $\varepsilon \in \mathbb{R}^+$ there is a $N \in \mathbb{Z}^+$ such that if $n > N$

$$(579) \quad |r_n^* - x_n| < \varepsilon$$

Since $\lim_n r_n^* = r$

$$(580) \quad \exists N' \in \mathbb{Z}^+ : \forall n > N' : |r_n^* - r| < \varepsilon$$

Let $M := \max\{N, N'\}$, if $n > M$, then for all $\varepsilon \in \mathbb{R}^+$

$$(581) \quad |r_n^* - x_n| < \frac{\varepsilon}{2}$$

$$(582) \quad |r_n^* - r| < \frac{\varepsilon}{2}$$

By the triangle identity

$$(583) \quad \begin{aligned} |x_n - r| &= |x_n - r_n^* + r_n^* - r| \\ &\leq |r_n^* - x_n| + |r_n^* - r| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon \end{aligned}$$

Hence for all $n > M$

$$(584) \quad |x_n - r| < \varepsilon$$

Or

(585)

$$\lim_n x_n = r$$

□